

Analyzing organic matter composition at intact biopore and crack surfaces by combining DRIFT spectroscopy and Pyrolysis-Field Ionization Mass Spectrometry[#]

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Abstract

In the clay-illuvial horizons (Bt) of Luvisols, surfaces of biopores and aggregates can be enriched in clay and organic matter (OM) relative to the bulk of the soil matrix. The OM composition of these coatings determines their bio-physico-chemical properties and is relevant for transport and transformation processes but is largely unknown at the molecular scale. The objective of this study was to improve the interpretation of spectra from Fourier transform infrared spectroscopy in diffuse reflectance mode (DRIFT) by using thermograms and released ion intensities obtained with pyrolysis-field ionization mass spectrometry (Py-FIMS) for a more detailed analysis of the mm-scale spatial distribution of OM components at intact structural surfaces. Samples were separated from earthworm burrow walls, crack coatings, uncoated cracks, root channels, and pin-hole fillings of the Bt-horizons of Luvisols. The information from Py-FI mass spectra enabled the assignment of OM functional groups also from spectral regions of overlapping DRIFT signal intensities to specific OM compound classes. In particular, bands from C=O and C=C bonds in the infrared range of wave numbers between 1,641 and 1,605 cm⁻¹ were related to heterocyclic N-compounds, benzonitrile, and naphthalene. The OM at earthworm burrow walls was composed of chemically labile aliphatic C-rich and rather stable lignin and alkylaromatic compounds whereas the OM of thick crack coatings and pinholes was dominated by heterocyclic N and nitriles and high-molecular compounds, likely originating from combustion residues. In combination with Py-FIMS, DRIFT applications to intact samples seem promising for generating a more detailed mm-scale spatial distribution of OM-related sorption and wettability properties of crack and biopore surfaces that may serve as preferential flow paths in structured soils.

Key words: soil organic matter / clay organic coatings / infrared spectroscopy / pyrolysis-field ionization mass spectrometry / preferential flow paths

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1 Introduction

Earthworm burrows, root channels, shrinkage cracks, and inter-aggregate spaces form a complex network in structured soils. During preferential flow events, mainly the surfaces of these macropores interact with reactive solutes and colloids, consisting of clay-organic complexes, dissolved organic matter, nutrients, and agrochemicals, in the percolating soil solution (Kodešova et al., 2008; Köhne et al., 2009). When considered as an interface between the porous soil matrix and the macropore domains in structured soils (e.g., Gerke, 2012), the structural surfaces represent “hot spots” with bio-physico-chemical properties that can be different from those of the bulk soil matrix (Nobles et al., 2003). Intact surfaces of biopores such as earthworm burrows (Jégou et al., 2001;

Don et al., 2008) and root channels (Moradi et al., 2012), and aggregate or crack coatings (Leue et al., 2013) can be enriched in organic matter (OM) with a composition that differs from that of the bulk soil (Ellerbrock and Gerke, 2004). Microbial activity (Bundt et al., 2001), OM adsorption (Guggenberger and Kaiser, 2003) and sorption of agro-chemicals (Stehouwer et al., 1994) were reported to be larger at side-walls of biopores than in bulk soil. The composition of OM at intact structural surfaces may influence preferential flow and mass transfer processes between macropores and soil matrix (Jarvis et al., 2007) by affecting the sorptivity and the wettability of surfaces along the flow paths (Ellerbrock and Gerke, 2013). Preferential flow and chemical leaching has been reported to occur in many structured soils (e.g., Jarvis et al.,

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2007) such as Luvisols developed from loess (e.g., Kodešova et al., 2011) and glacial till in subsurface drained arable fields (e.g., Gerke and Köhne, 2004; Wichtmann et al., 1998).

Luvisols are characterized by a vertical differentiation in clay contents, which result from clay migration. In the clay-illuvial horizon (Bt; WRB classification: argillic horizon; WRB, 2006), clay and clay-OM complexes are preferably located at surfaces of inter-aggregate pores and larger pores (Kodešova et al., 2009; 2012). The mm-scale spatial distribution of clay-organic coatings is reflected in content and composition of OM, in particular by surface-type specific proportions of aliphatic and carboxylic functional OM groups observed for intact subsoil samples (including Luvisols) from soils of different parent material and land use (Leue et al., 2010a; Leue et al., 2013).

The OM composition can be characterized by Fourier transform infrared (FTIR) spectroscopy (Smith, 1995; Demyan et al., 2012) in the mid-infrared range (MIR; wave lengths of 2.5 to 25 μm) with respect to the relative contents of functional groups such as C–H, N–H, C=O, and C=C groups. In the diffuse reflectance infrared Fourier transform (DRIFT) mode, the technique can be applied to homogenized and diluted samples as well as to intact soil samples (Haberhauer and Gerzabek, 1999; Ellerbrock et al., 2009). However, the identification of OM functional groups by means of DRIFT spectroscopy in soil samples is limited by overlapping bands of different organic and mineral functional groups. Statistical methods, e.g., partial least square regression (PLSR), have been applied to relate combinations of DRIFT signal intensities to specific soil parameters such as organic carbon (OC) and other elemental contents (e.g., Viscarra-Rossel et al., 2006) and to OM functional groups measured by nuclear magnetic resonance (NMR) spectroscopy (Leifeld, 2006; Ludwig et al., 2008). The complementary use of pyrolysis-field ionization mass spectrometry (Py-FIMS) and FTIR/DRIFT spectroscopy was suggested as a potential tool to characterize OM composition (Smidt et al., 2005). However, these complementary techniques have not yet been applied to samples from intact structural surfaces of soils, and a comprehensive characterization of OM at those structures is still missing.

Py-FIMS offers information on OM composition in soils by identifying and quantifying important classes of chemical compounds in mass spectra (Schulten and Leinweber, 1999). Molecular information on OM constituents, using specific molecular marker signals, is obtained after thermal degradation, soft ionization in a high electric field, and mass-spectrometric detection of the ionized molecules (Leinweber

and Schulten, 1999). The mass-spectrometric analysis allows differentiating between masses from m/z 15 up to 900. However, in this range the low masses m/z 15...56 (low molecular weight compounds) are not assigned to a specific naturally occurring compound class because these molecules can be produced by various thermal decomposition reactions (Leinweber et al., 2009a). Thermograms of the volatilization of OM constituents allow concluding on chemical bonds between OM and soil minerals and within the organic molecules. The combination of mass spectra and thermograms for the bulk OM, OM compound classes, or single ions allows a detailed characterization of the OM composition beyond the possibilities of DRIFT spectroscopy. However, Py-FIMS leaves the sample surfaces under study not intact and the measurements are more costly and time-consuming than DRIFT spectroscopy. Thus, the combination of these two methods likely reveals differences in OM composition at intact structural surfaces in more detail than a single method can. In particular, the Py-FIMS information on the different OM compound classes and thermal stabilities, the latter indicating organic-mineral bonds, may complement the non-destructive DRIFT spectroscopic OM characterization.

The objective of this study was to improve the interpretation of the DRIFT signal intensities by complementary thermograms and ion intensities from Py-FIMS, and achieve detailed characterization of the mm-scale spatial distribution of OM components at intact structural surfaces by combining data from these two methods. In this first step, we used sample material separated from biopore and crack surfaces from Bt-horizons of Luvisols. We compared samples from Luvisols developed from loess and glacial till to consider effects of different soil parent material.

2 Material and methods

2.1 Soils and sampling

Soil samples were collected from a Haplic Luvisol developed from loess (L) at an experimental site located in the N of the Czech Republic (Hnevceves, near Hradec Kralove); the other samples were from a Haplic Luvisol developed from glacial till (T), located in NE Germany (Holzendorf, near Prenzlau). The basic soil and site information are given in Table 1. Both subsoil Bt-horizons were well-structured with cracks, fissures, and earthworm burrows, which could potentially serve as preferential flow paths (for the loess soil at Hnevceves see details in Kodešova et al., 2011; Fér and Kodešova, 2012).

Table 1: General information on the sampling sites and properties of the two investigated Haplic Luvisols (WRB, 2006). Mean of three replicates. MAP: mean annual precipitation; MAT: mean annual temperature; SOC: soil organic carbon.

Site			Bt-horizon										
Site name	Longitude / °E	Latitude / °N	Altitude / m asl	MAP / mm y ⁻¹	MAT / °C	Parent material	Land use	Depth / cm	pH (CaCl ₂)	Clay / %	Silt / %	Sand / %	SOC / %
Hnevceves	15°43'03"	50°18'47"	273	618	8.5	loess (L)	arable	60–80	6.6	34.6	56.2	9.2	0.44
Holzendorf	13°47'11"	53°22'45"	48	501	8.7	glacial till (T)	arable	40–60	6.4	16.5	35.0	48.5	0.31

At the two sites, soil pits of edge lengths of 120 cm x 120 cm were excavated, from where 12 larger soil blocks of approx. 15 cm height, 20 cm length, and 30 cm width were cut out of the Bt-horizon using a spade. The blocks were immediately packed in tinfoil to prevent from evaporation or cracking (Leue et al., 2013). In addition, disturbed soil samples were taken for textural and chemical analyses (Table 1). In the laboratory, about 30 smaller sub-samples of about 5 to 10 cm edge length were manually separated from the larger soil blocks of each site to obtain intact crack and biopore surfaces. Six different surface types were identified at the sub-samples by visual inspection: uncoated cracks (CS-C), relatively thin (*i.e.*, bright) crack coatings (CS±C), thick crack coatings (CS+C), pinhole fillings (PIN), earthworm burrows (EB), and root channels (RC; Table 2). From the sub-samples, the outermost layer of a thickness of < 1 mm was manually separated from the underlying soil matrix for each identified surface type area, using a knife and a scalpel. Hence, the separated material of each surface type represents a mixed sample of this surface type from one Bt-horizon (*i.e.*, soil pit). The separated material with masses between 5 and 10 g of each surface type was dried over silica gel in a desiccator.

2.2 Diffuse reflectance infrared Fourier transform (DRIFT) spectroscopy

For DRIFT spectral analyses, 1 mg soil material was mixed with 99 mg potassium bromide (three sample repetitions), finely ground in an agate mortar, and poured into standard DRIFT cups. The DRIFT spectra were recorded as 16 co-

added scans between wave number (WN) 4,000 and 400 cm⁻¹ at a resolution of 1 cm⁻¹, and related to a background spectrum of a gold target (99% Infragold). The spectra were converted to Kubelka–Munk (KM) units (Kubelka, 1948), smoothed (boxcar moving average algorithm, factor 25), and corrected for baseline shifts using the software WIN-IR Pro 3.4 (Digilab, MA, USA). Each DRIFT spectrum was analyzed for specific signal intensities of organic and mineral matter, using band assignments given in literature (see Supplemental material 1). The signal intensities of the bands were measured as heights from the total baseline with the exception of the N–H bands (WN 3,219 and 3,123 cm⁻¹) and the C–H band intensities (WN 3,020, 2,956, 2,926, and 2,856 cm⁻¹), which were measured as the vertical distance from a local baseline plotted between tangential points to consider the effect of the broad O–H band at WN 3,435 cm⁻¹. The signal intensities of the single OM-dominated DRIFT bands at WN 3,219, 3,123, 3,020, 2,956, 2,926, 2,856, 1,738, 1,722, 1,712, 1,688, 1,641, 1,624, 1,616, 1,605, 1,565, 1,547, 1,529, 1,514, 1,461, 1,425, 1,400, and 918 cm⁻¹ were related to (*i.e.*, divided by) the summed signal intensities of all these bands to consider differences in sample- and site-specific DRIFT signal intensity levels caused by different OM contents and soil parent material of the samples (Table 2). The signal intensities of the bands at WN 3,690, 3,616, 3,435, 1,983, 1,871, 1,788, 1,094, 1,031, 790, and 693 cm⁻¹ were used without normalization against the summed OM signals since these bands were dominated by vibrations of functional groups from soil minerals (O–H, Si–O).

Table 2: Separated structural surface materials, sample nomenclature, and description. Mean contents of organic C, N, C/N ratio, and C–H/C=O ratio; *n* = 3.

Sample	Surface type description	C /%	N /%	C/N ratio	C–H/C=O ratio
Loess (L)					
CS-C(L)	bright crack surfaces without coatings, as far as visible	0.39	0.08	4.8	0.034
CS±C(L)	visible but not very distinctive (<i>i.e.</i> , thin) clay-organic coatings at crack surfaces, colored more brownish than CS-C	0.63	0.12	5.8	0.035
CS+C(L)	very distinctive (<i>i.e.</i> , thick) clay-organic coatings at crack surfaces, dark brown color	1.42	0.21	6.9	0.027
PIN(L)	clay-organic fillings of tapped blind holes, <i>i.e.</i> , cylindrical pinhole pores	1.44	0.21	6.9	0.026
RC(L)	root channel with clay-organic fillings, found only in the Bt-horizon of the loess-derived Luvisol	1.34	0.19	6.9	0.034
EB(L)	earthworm burrows, colored more brownish than the soil matrix and more grayish than the coated cracks	0.81	0.13	6.3	0.070
Glacial till (T)					
CS-C(T)	see CS-C(L)	0.24	0.06	3.7	0.085
CS±C(T)	see CS±C(L)	0.26	0.07	3.8	0.067
CS+C(T)	see CS+C(L)	0.35	0.08	4.5	0.042
PIN(T)	see PIN(L)	0.77	0.16	4.8	0.034
PINb(T)	black colored pinhole filling, darker than PIN(T)	0.34	0.08	4.0	0.026
EB(T)	see EB(L)	0.64	0.11	5.9	0.078

2.3 Pyrolysis-field ionization mass spectrometry (Py-FIMS)

From the ground and homogenized samples used for DRIFT spectroscopy, aliquots of 5 mg (3 repetitions) were lyophilized and thermally degraded in the ion source (emitter: 4.7 kV, counter electrode −5.5 kV) of a double-focusing Finnigan MAT 95 mass spectrometer. The samples were heated over a time period of 15 minutes in a vacuum of 10^{-4} Pa from 50°C to 650°C, in temperature steps of 10°C. Between magnetic scans the emitter was flash heated to avoid residues of pyrolysis products. For each sample, 65 spectra were recorded for the mass range 15 to 900 m/z . The ion intensities were referred to 1 mg of the sample. Volatile matter (VM) was calculated as mass loss in percentage of sample weight. For data interpretation, marker signals (m/z) that are assigned to relevant OM compound classes were used according to *Hempling and Schulten* (1988), *Schnitzer and Schulten* (1992), *Schulten and Leinweber* (1996), *van Bochove et al.* (1996), and *Leinweber et al.* (2009b; 2013). The ion intensities of the marker signals were attributed to the following compound classes (*Leinweber and Schulten*, 1999): CHYDR—carbohydrates, PHLM—phenols and lignin monomers, LDIM—lignin dimers, LIPID—lipids, alkenes, fatty acids, and *n*-alkyl esters, ALKY—alkylaromatics, NCOMP—heterocyclic N and nitriles, STERO—sterols, PEPTI—amides (amino acids, peptides, aminosugars), SUBER—suberin, FATTY—free fatty acids ($n\text{-C}_{16}$ to $n\text{-C}_{34}$), m/z 15...56 (low molecular weight compounds), and M+H (protonated molecules) and ^{13}C peaks. The ion intensities of benzonitrile and naphthalene were used as a parameter for highly stable OM components. During the pyrolysis, no relevant amounts of benzonitrile and naphthalene were formed (e.g., *Kiersch et al.*, 2012a). The total ion intensity (TII) of each compound class was normalized by the summed TII measured for all compound classes to obtain relative values (% TII).

2.4 Statistical analyses

For the samples of each Luvisol ($n = 6$), the normalized DRIFT signal intensities (KM units) were correlated with the normalized TII values obtained by Py-FIMS using the Spearman correlation. For the given degrees of freedom ($df = 4$), the correla-

tion coefficient (r) was significant at the level of 5 % for $r \geq 0.81$ and at the level of 1 % for $r \geq 0.92$. In case all samples ($n = 12$; $df = 10$) were considered, r was significant at the level of 5 % for $r \geq 0.58$ and at the level of 1 % for $r \geq 0.71$. The discriminant analyses of the 12 samples were performed by a partial least squares regression (PLSR) using R, Version 3.1.1 (*R Core Team*, 2014) with module *p/s* (SIMPLS, cross-validation: leave-one-out). Larger absolute loading values of signal intensities in certain WN regions imply a greater importance of these WN for the cumulated values of the principal component 1 or 2 displayed in the discriminant plot.

3 Results

3.1 DRIFT spectroscopy

The DRIFT spectra of the separated surface material from the loess- and the till-derived Bt-horizons can be found in Supplemental material 2. The discriminant analysis of the DRIFT spectra (Fig. 1), used for a first overview on differences in OM and mineral composition, revealed two groups among the loess (L) samples: (1) uncoated cracks (CS-C), thin crack coatings (CS±C), and earthworm burrows (EB), and (2) thick crack coatings (CS+C), pinholes (PIN), and root channels (RC). These groups differed with respect to component 2, mainly caused by negative loading values around WN 2,200, 1,900, 1,800, 1,550, and 1,200 cm^{-1} (Fig. 1c). Among the loess samples, the greatest difference occurred between uncoated cracks and thick crack coatings (and root channels). Among the till (T) samples, pinholes differed from the other samples in terms of component 1, based on different loading values between WN 2,000 and 1,800 cm^{-1} (Fig. 1b). Considering component 1 for all samples (from both Luvisols), the greatest differences were found between pinholes and earthworm burrows while high similarities occurred between DRIFT spectra of earthworm burrows and the corresponding uncoated cracks.

The earthworm burrows of both Luvisols showed highest summed DRIFT signal intensities of WN 2,956, 2,926, and

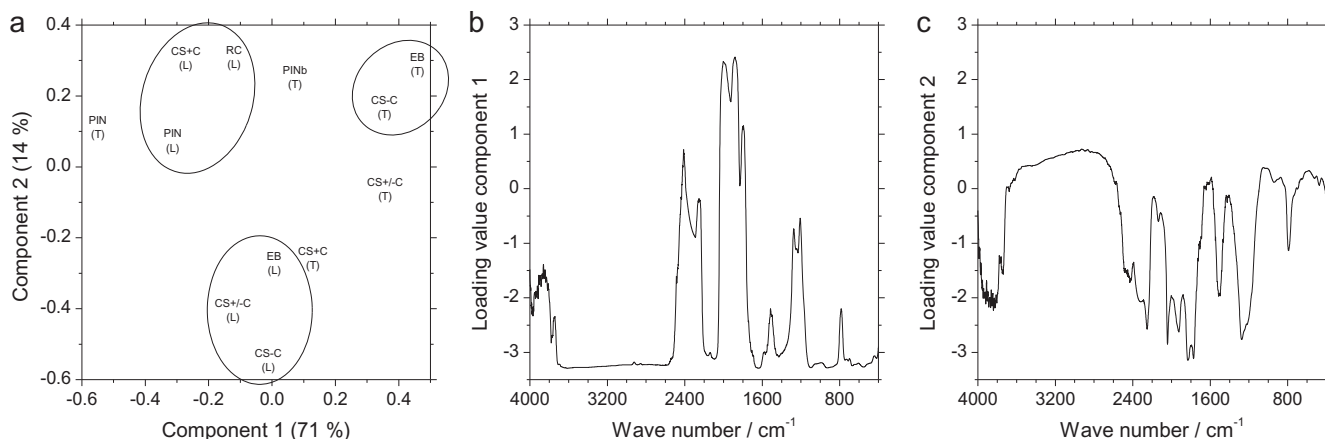


Figure 1: (a) Discriminant analysis of the DRIFT spectra of samples separated from structural surfaces (CS-C: uncoated cracks; CS±C: thin crack coatings; CS+C: thick crack coatings; PIN: pinhole fillings; RC: root channel; EB: earthworm burrows) separated from Bt-horizons of a loess- (L) and a till-derived (T) Luvisol. Loading values of the DRIFT signal intensities of component 1 (b) and component 2 (c); higher absolute loading values imply higher importance of the WN region (*i.e.*, the DRIFT signal intensities) on the components.

2,856 cm^{-1} , assigned to aliphatic C–H stretching (Baes and Bloom, 1989; Capriel et al., 1995; Capriel, 1997); the level of the summed signal intensities was slightly higher for the till samples as compared to the loess-derived ones (Fig. 2). The summed C–H signal intensities were found to be highest for earthworm burrows of both sites. The ratio between the signals of C–H groups at WN 2956 and 2926 cm^{-1} and that of C=O groups at WN 1,712 and 1,605 cm^{-1} (Ellerbrock et al., 2005) was similar for the loess samples, except for the earthworm burrows. In contrast, for the till samples the C–H/C=O ratio was higher for uncoated cracks and thin crack coatings as compared to thick crack coatings and pinholes (Table 2). The C–H/C=O ratios of the earthworm burrows from both sites were found to be higher compared to those of the corresponding uncoated cracks and crack coatings.

Among the loess samples the summed signal intensities of WN 1,738, 1,722, and 1,712 cm^{-1} , assigned to C=O stretching in lactones, ketones, aldehydes, and fatty acids (Hesse et al., 1984; Stevenson, 1994; Senesi et al., 2003) were higher for uncoated cracks and thin crack coatings and earthworm burrows, compared to thick crack coatings, pinholes, and root channels (Fig. 2). For the till samples the signals were decreased only for pinholes as compared to uncoated cracks

and earthworm burrows. For both the till and loess samples, the summed signal intensities of WN 1,641, 1,624, 1,616, and 1,605 cm^{-1} , assigned to C=C, C–N, C–O, and N–H groups (Baes and Bloom, 1989; Stevenson, 1994; Senesi et al., 2003; Bornemann et al., 2008), were highest for thick crack coatings, root channels, and pinholes, while lower and similar levels were found for earthworm burrows, uncoated cracks, and thin crack coatings (Fig. 2). The summed signal intensities at WN 1,547, 1,529, and 1,514 cm^{-1} , assigned to C=C groups of aromatic compounds and N–H groups of amides (Baes and Bloom, 1989; Senesi et al., 2003; Bornemann et al., 2008), were found to be increased for earthworm burrows from both Luvisols. For the loess samples, the signals were smaller for thick crack coatings, pinholes, and root channels, compared to uncoated cracks and thin crack coatings. For the till samples, the smallest values were found for the pinholes and uncoated cracks. The signal intensities at WN 1,400 cm^{-1} , assigned to C=O, O–H, and N–O groups (Hesse et al., 1984; Senesi et al., 2003; Spence and Kelleher, 2012), were found to be highest for earthworm burrows from both sites. For the loess samples, the signals increased in the order CS–C < CS±C < (CS+C, PIN, RC), which was in contrast to the till samples, which showed smallest values for the pinholes.

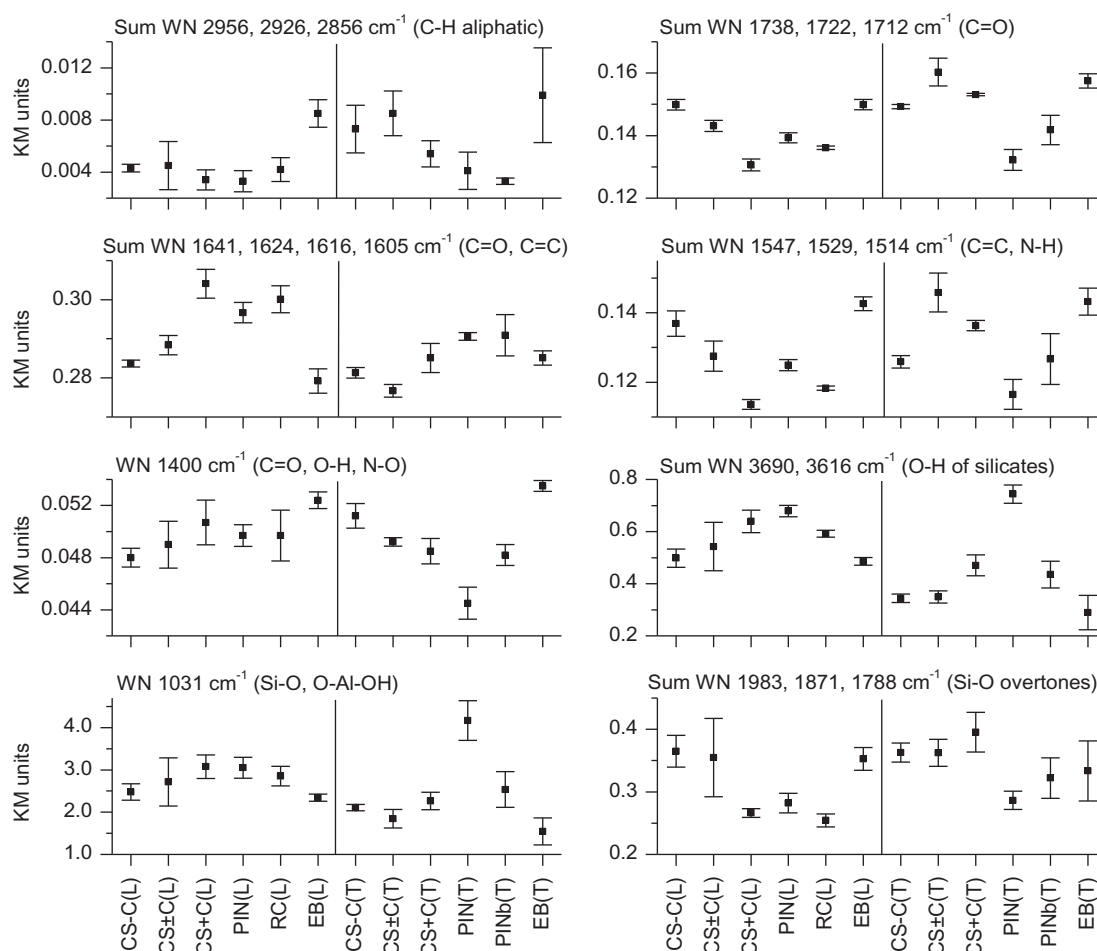


Figure 2: Signal intensities in DRIFT spectra of samples separated from structural surfaces (CS–C: uncoated cracks; CS±C: thin crack coatings; CS+C: thick crack coatings; PIN: pinhole fillings; RC: root channel; EB: earthworm burrows) separated from Bt-horizons of a loess- (L) and a till-derived (T) Luvisol (mean and standard deviation).

The summed DRIFT signal intensities at WN 3,690 and 3,616 cm^{-1} , assigned to the O–H stretching of structural hydroxyls of clay minerals (van der Marel and Beutelspacher, 1976; Madejova and Komadel, 2001) as well as the signal intensity at WN 1031 cm^{-1} , assigned primarily to Si–O and O–Al–OH groups from silicates, were higher for the loess samples than for the till-derived ones, except for the till-derived pinholes, which showed the highest signal intensities among all samples (Fig. 2). For both Bt-horizons, the summed signal intensities of the silicate bands decreased in the order $\text{PIN} > \text{CS}+\text{C} > (\text{RC}, \text{PINb}) \geq (\text{CS}-\text{C}, \text{EB}, \text{CS}\pm\text{C})$. The summed signal intensities of the ‘bulk mode’ bands (P     et al., 2001), i.e., the combination or overtones of symmetric and asymmetric SiO_2 bands (Benesi and Jones, 1959) at WN 1,983, 1,871, and 1,788 cm^{-1} were found to be higher for uncoated cracks, thin crack coatings, and earthworm burrows as compared to thick crack coatings, pinholes, and root channels of the loess samples (Fig. 2). Amongst the till samples, lower signal levels were found for pinholes and earthworm burrows, compared to the uncoated cracks and crack coatings.

3.2 Pyrolysis-field ionization mass spectrometry

The Py-FI mass spectra (Supplemental material 3) of most samples were dominated by intense signals of nitrogen-containing compounds at m/z 58, 67 (pyrrole), 79 (pyridine), 81 (methylpyrrole), 93 (methyl-pyridine), 103 (benzonitrile), and of naphthalene (m/z 128), and lignin monomers (m/z 156, 168, 180, and 194). The till samples additionally revealed prominent signals of C_{18} to C_{20} -alkenes at m/z 252, 266 and 280 and, particularly for the uncoated cracks (T), intensive signals of C_{15} to C_{17} -alkanes at m/z 212, 226, and 240. Mass spectra from earthworm burrows showed intensive signals of carbohydrates at m/z 96, 110, and 132, and of indole (m/z 117). All thermograms (Supplemental material 3, inserts upper right) of the samples from structural surfaces illustrated the thermal release of substances at temperatures mainly $> 400^\circ\text{C}$.

The discriminant analysis of the Py-FI mass spectra showed similarities between two groups of loess samples: (1) un-

coated cracks, thin crack coatings, and earthworm burrows, and (2) thick crack coatings, pinholes, and root channels (Fig. 3a). The differences between these groups of loess samples were small compared to the differences between the till samples, which showed similarities only between thin and thick crack coatings. With respect to component 1 the greatest difference was obvious between till-derived earthworm burrows and pinholes. The Py-FI mass spectra of earthworm burrows from both Luvisols were relatively similar, largely caused by m/z signals from polysaccharides, such as the strongly negative loading values from furaldehydes of m/z 96 and 110 (Fig. 3b). Considering principal component 2, the till-derived pinholes differed extremely from the rest of the samples caused by extremely high signals of m/z 103 (ethannitrile and benzonitrile), besides signals of m/z 78 (benzene) and 128 (naphthalene) as the most important loading values (Fig. 3c). For both principal components, low molecular weight compounds (m/z 15...56) accounted for differences between the samples.

The assignment of marker signals to compound classes (Fig. 4) indicated that the proportions of carbohydrates (CHYDR) in the loess samples were two times larger than in the till samples, except for the till-derived earthworm burrows. Irrespective of the soil parent material, the earthworm burrows contained larger CHYDR proportions and the largest proportions of phenols and lignin monomers (PHLM) compared to the other structural surfaces. The loess-derived thin and thick crack coatings as well as the pinholes had slightly higher PHLM proportions than the corresponding till samples. The till-derived cracks revealed higher PHLM proportions than the till-derived thin and thick crack coatings and showed the release of volatile PHLM compounds at temperatures $< 100^\circ\text{C}$ (Fig. 5). For the loess-derived uncoated cracks, thick crack coatings, and pinholes an additional PHLM release peak at 450°C was observed.

The proportions of lignin dimers (LIDM) were generally low and decreased in the order $\text{CS}-\text{C} > \text{CS}\pm\text{C} > (\text{CS}+\text{C}, \text{PIN})$ for both Luvisols (Fig. 4). The LIDM proportions of earthworm burrows were higher than those of pinholes and of the till-derived uncoated cracks and crack coatings. The proportions of

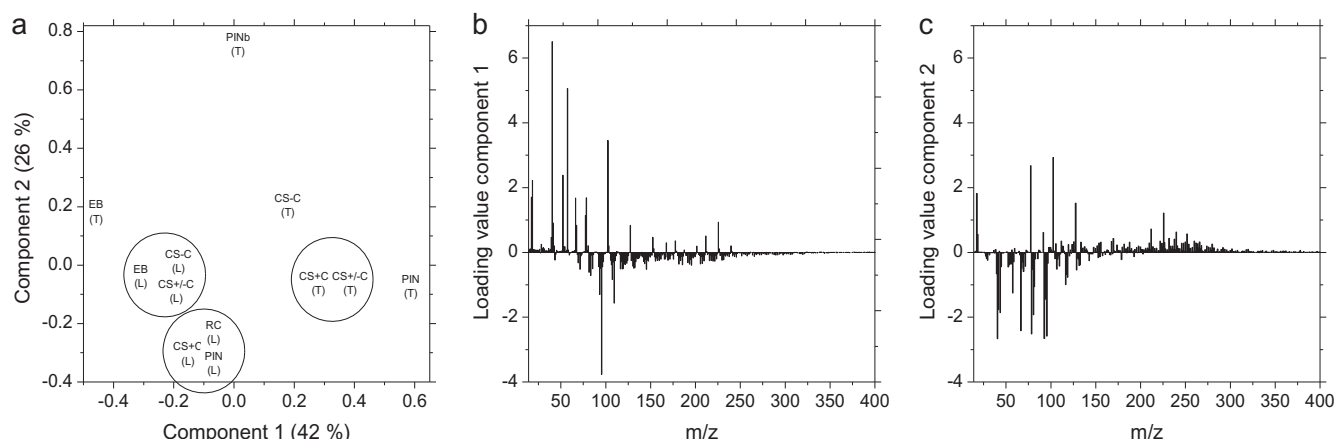


Figure 3: (a) Discriminant analysis of the Py-FIMS m/z values of samples separated from structural surfaces (CS-C: uncoated cracks; CS±C: thin crack coatings; CS+C: thick crack coatings; PIN: pinhole fillings; RC: root channel; EB: earthworm burrows) separated from Bt-horizons of a loess- (L) and a till-derived (T) Luvisol. Loading values of the m/z numbers of component 1 (b) and component 2 (c).

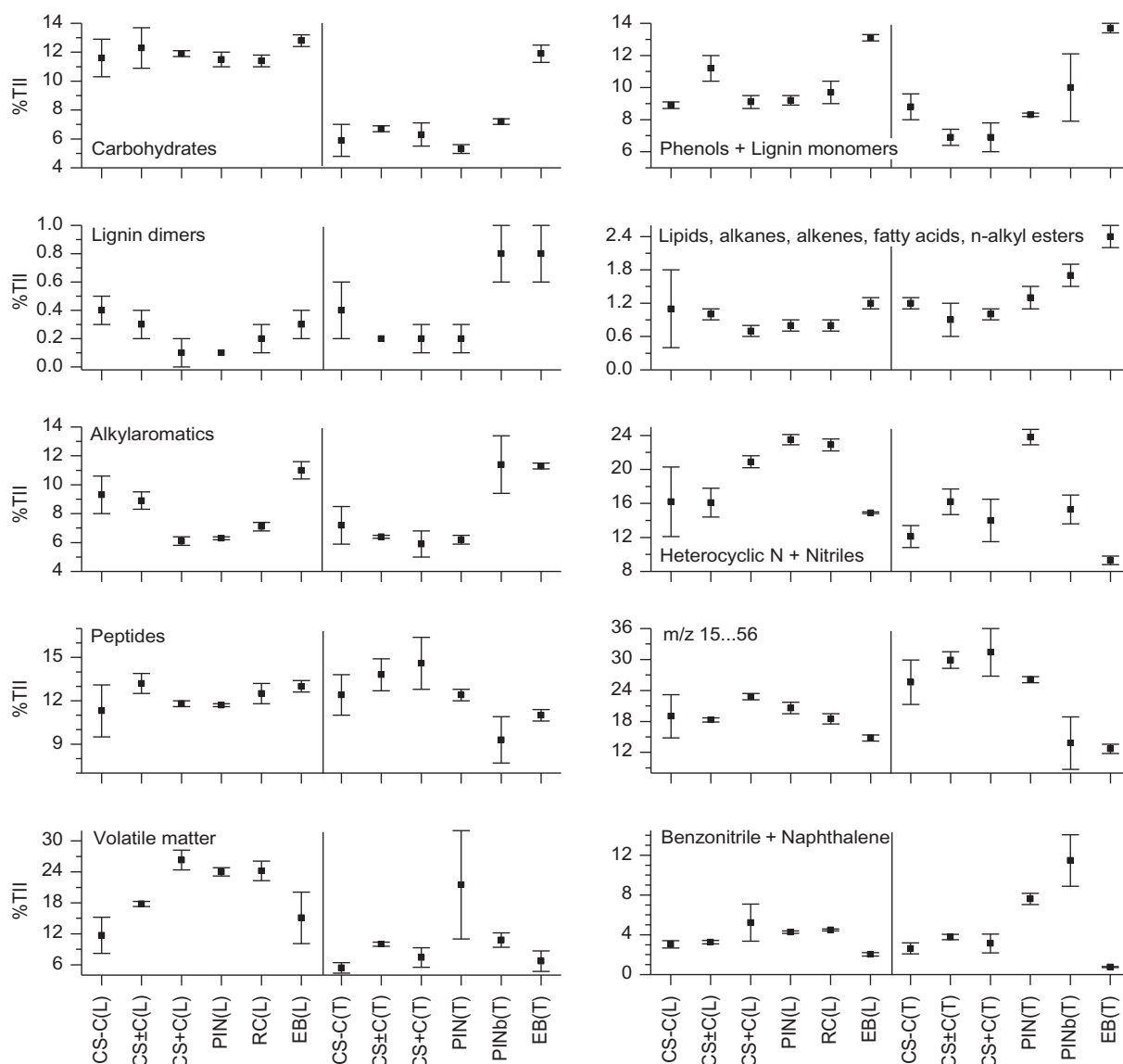


Figure 4: Organic matter (OM) compound classes as percentage of total ion intensity (% TII) determined from Py-FI mass spectra of samples separated from structural surfaces (CS-C: uncoated cracks; CS±C: thin crack coatings; CS+C: thick crack coatings; PIN: pinhole fillings; RC: root channel; EB: earthworm burrows) separated from Bt-horizons of a loess- (L) and a till-derived (T) Luvisol (mean and standard deviation).

lipids, alkanes, alkenes, fatty acids, and n-alkyl esters (LIPID) were generally low, with larger values for the earthworm burrows of both sites compared to the other surface types. In the thermograms of LIPID (Fig. 5), a first release peak near 100°C was found for uncoated cracks from both sites and for till-derived thick crack coatings, corresponding to increased signals from alkanes attributed to diesel residues (see below). The till-derived earthworm burrows released a maximum of LIPID compounds at around 500°C compared to a maximum release from the other samples at about 550°C. Alkylaromatics (ALKY) decreased in the order EB > CS-C > PIN > CS+C for both Luvisols. Corresponding to the release of LIPID, a first ALKY release peak near 100°C in case of uncoated cracks from both sites and of till-derived thick crack coatings was observed (Fig. 5).

Heterocyclic N compounds and nitriles (NCOMP) ranged between 9 and 24% TII (Fig. 4), decreasing in the order PIN > CS+C > CS-C > EB for both Luvisols. In case of the till samples, higher NCOMP releasing temperatures of pinholes, compared to those of earthworm burrows, were found (Fig. 5). For the proportions of amino acids, peptides, and aminosugars (PEPTI) no clear trend between biopores and cracks was observed (Fig. 4). The lowest proportions of low-molecular weight compounds (m/z 15...56) were found for earthworm burrows from both Luvisols, in contrast to large values of the cracks and pinholes. The release of VM was generally smaller for the till than for the loess samples. For both Luvisols, relatively more VM was released from thick crack coatings and pinholes than from uncoated cracks and earthworm burrows; the latter revealed the same VM levels. The proportions of sterols (STERO), suberin (SUBER), and free fatty acids (FATTY) were < 0.1% TII in most cases and

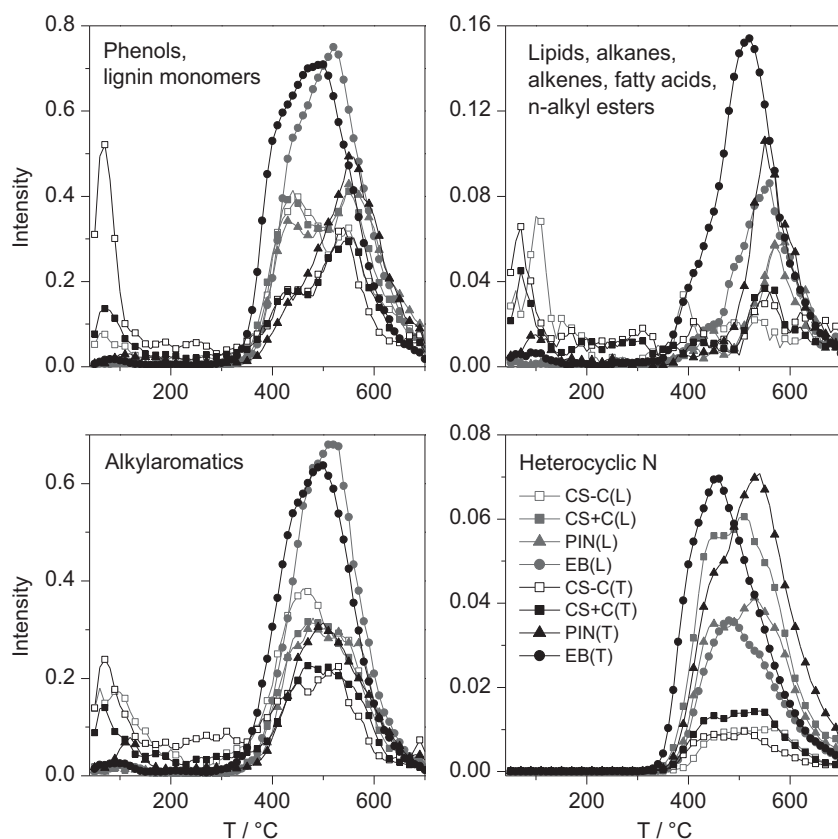


Figure 5: Py-FIMS thermograms of OM compound classes from selected samples of the loess- (L) and till-derived (T) Bt-horizon.

not further discussed here. The summed ion intensities of benzonitrile and naphthalene (Bn+Na) ranged between 0.8 and 11.5 % TII (Fig. 4) with the lowest values for earthworm burrows and the largest ones for thick crack coatings and pinholes. For the till-derived pinholes, the total released Bn+Na values were extremely large (not shown).

4 Discussion

4.1 Comparison of DRIFT and Py-FIMS results

The discriminant analyses of both the DRIFT and Py-FIMS data revealed a similar clustering for the same two sample groups from the loess-Bt on the one hand and a picture of heterogeneity in the OM composition among the till samples on the other (Figs. 1a and 3a). In contrast to the loess samples, many correlations between the DRIFT signal intensities and the %TII obtained by Py-FIMS were not significant for the till samples (Table 3) and, thus, could not be considered here. However, also for the loess samples some relations between DRIFT and Py-FIMS, which were expected on the basis of theoretical considerations, could not be proven possibly due to the small number of samples ($n = 12$) and to the heterogeneity of the samples from different structural surfaces types and different Luvisol parent materials. The samples from the till-derived Luvisol revealed a higher heterogeneity in OM and mineral composition, especially between the coarser textured uncoated cracks and the finer textured pinhole fillings with high DRIFT signal intensities from soil silicates (WN 3,690, 3,616, and 1,031 cm^{-1} ; Fig. 2). Although the till-derived

Luvisol had a relatively coarse texture (Table 1), crack coatings and pinhole fillings in the Bt-horizon were enriched in finely textured clay material, resulting from clay migration. Hence, the samples separated from the till-Bt structures showed a wide range in texture. This range was smaller for the loess-derived samples since this Luvisol showed a generally finer texture.

The bulk mode bands at WN 1,983, 1,871, and 1,788 cm^{-1} are known to increase with the particle size of the samples (Leue et al., 2010b); higher bulk mode bands indicate coarser particles (sand fraction). The high loading values of this spectral region (Fig. 1) confirm that the sample texture affected the characteristics of DRIFT spectra (although the samples were finely ground), an effect that finally may have contributed to the differences in signal intensities from DRIFT and Py-FIMS. The negative correlations between the DRIFT bulk mode bands and the values of NCOMP, Bn+Na, C, N, and VM (Table 3) underlined that the OM contents of the sample as well as the NCOMP proportions decrease with increasing particle size. The positive correlations between the DRIFT signals of WN 1,641 to 1,605 cm^{-1} and NCOMP, Bn+Na, C, N, and VM confirmed that, firstly, these signal intensities could be assigned to heterocyclic N compounds, benzonitrile and naphthalene. Secondly, these DRIFT signal intensities seemed useful to predict C and N contents of soil samples. Thirdly, the WN range includes signal intensities from O–H bending of water molecules in hydration layers of phyllosilicates (Mendelovici et al., 1995); the evaporation of water during the pyrolysis resulted in increased VM values. In general, the DRIFT signals from O–H groups and the proportions of

VM and low-molecular weight compounds (m/z 15...56) had to be considered as auto-correlated.

The DRIFT signal intensities of aliphatic C–H groups (WN 2,956, 2,926, and 2,856 cm^{-1}) were correlated significantly with carbohydrates, phenols and lignin monomers, lipids, and alkylaromatics in the loess samples, in accordance to the theoretical considerations. While the correlations between the

DRIFT signals at WN 1,738, 1,722, 1,712, and 1,688 cm^{-1} (dominated by C=O groups) and lipid proportions were plausible, the correlations of DRIFT signals between WN 1,983 and 1,688 cm^{-1} with the proportions of lignin dimers, alkylaromatics, and in part with lipids, could not be explained by theoretical considerations. The DRIFT signal intensities at the bands of WN 1,641, 1,624, and 1,616 cm^{-1} were significantly correlated with the NCOMP proportions determined by Py-

Table 3: Correlations between DRIFT signal intensities (normalized KM values) and Py-FIMS compounds (%TII) for the loess (L), till (T) samples, and both, loess and till samples (\$). Only significant r values were displayed; a minus (“–”) indicates significantly negative correlations. Levels of significance: 5 % (L, T, \$) and 1 % (LL, TT, \$\$). Bn+Na: benzonitrile and naphthalene. Grey-shaded fields: correlation expected from theoretical considerations.

Band No.	WN / cm^{-1}	CHYDR	PHLM	LDIM	LIPID	ALKY	NCOMP	m/z 15...56	Bn+Na	C / %	N / %	VM
1	3,690						LL, T, \$\$		LL	L, \$\$	L	L, T
2	3,616						LL, T, \$\$	L	LL	\$	L	L, T
3	3,435						L, \$\$	L	LL, T, \$\$			L
4	3,219									L	L	
5	3,123											
6	3,020											
7	2,956	L	L		L	L						
8	2,926		L			L						
9	2,856	L	LL			L						
10	2,132											
11	1,983			L	L	L	–LL, –\$\$		–L	–LL, –\$\$	–LL, –T, –\$\$	–LL, –TT, –\$\$
12	1,871			L	LL	L	–LL, –\$\$		–L	–LL, –\$\$	–LL, –T, –\$\$	–LL, –TT, –\$\$
13	1,788			L	L	L	–L		–L	–L, –\$\$	–LL, –\$\$	–LL, –TT, –\$
14	1,738			L	LL	L						
15	1,722			L	LL	LL						
16	1,712			L	LL	L						
17	1,688			L	LL	L						
18	1,641						LL, T, \$\$		LL	L, \$\$	L, T, \$\$	LL, T, \$\$
19	1,624						L, \$\$	L	LL	\$\$	L, \$\$	L, \$\$
20	1,616						L	L	LL	L, \$\$	L, \$\$	LL, \$
21	1,605								LL	L	L	LL
22	1,565	\$										
23	1,547				L							
24	1,529			L	LL	LL						
25	1,514			L	LL	LL						
26	1,461				LL	LL						
27	1,425	\$	\$									
28	1,400	T	\$									
29	1,094						TT, \$\$	L			\$	TT, \$\$
30	1,031						L, TT, \$\$	L	LL	L	L, \$	L, TT, \$\$
31	918						T, \$\$	L	LL, \$			\$
32	790						T					T
33	693						T, \$\$	L	\$			TT, \$

FIMS (Table 3). While the band at $\text{WN } 1624\text{ cm}^{-1}$ could be assigned to N-containing functional groups (amongst other groups; Stevenson, 1994), the bands at $\text{WN } 1,641$ and $1,616\text{ cm}^{-1}$ could not be assigned to such groups by the help of literature references. These discrepancies showed that, although the agreement between results from DRIFT and Py-FIMS was high in some cases, an unambiguous assignment of functional groups (DRIFT) to compound classes (Py-FIMS) and vice-versa was not possible in every case.

The strong correlation between the DRIFT signal intensities of the bands at $\text{WN } 1,641$, $1,624$, $1,616$, and $1,605\text{ cm}^{-1}$ and the Py-FIMS signals of Bn+Na, as found for the loess samples (Table 3; Supplemental material 4; $R^2 = 0.93$), can be assigned to the aromatic C=C stretching of these chemically stable OM compounds. For the till-derived pinholes, the extremely large detected Bn+Na amounts probably represented only a small proportion of the total Bn+Na contained in these samples. Due to intensive aggregation with soil particles, Bn+Na could not be volatilized completely during the pyrolysis and remained as part of the ash in the sample holder (data not shown). Thus, particularly for the pinhole fillings of the till-derived Luvisol, the Bn+Na contents were strongly underestimated.

4.2 Earthworm burrow walls

In comparison to the other structural surface types, the OM of earthworm burrow walls was enriched in aliphatic and aromatic C–H groups and aromatic C=C groups (Fig. 2), which corresponded to increased proportions of CHYDR, PHLM, ALKY, NCOMP, PEPTI, and m/z 15...56 (Fig. 4). The OM composition of the earthworm burrows from both Luvisols was relatively similar, reflecting the dominance of undecomposed or poorly decomposed material incorporated into the earthworm burrow walls. The input of “fresh” material into the biopores can be seen as the main reason for the dissimilarities to the OM composition of thick crack coatings and pinhole fillings.

The ratio between the signals of the aliphatic C–H groups ($\text{WN } 2,956$ and $2,926\text{ cm}^{-1}$) and that of C=O groups at $\text{WN } 1,712$ and $1,605\text{ cm}^{-1}$ can be used to characterize the OM composition and as a proxy for the potential wettability of OM at intact structural surfaces (Ellerbrock et al., 2005; Leue et al., 2013). The increased C–H/C=O ratios of the earthworm burrows from both sites, compared to the corresponding uncoated cracks (Table 2), were in accordance to DRIFT mapping results obtained from intact sample surfaces investigated by Leue et al. (2010a; 2013). The proportions of aliphatic C / carbohydrates (Figs. 2 and 4) in combination with volatilization maxima at temperatures $> 400^\circ\text{C}$ (Figs. 5 and Supplemental material 3) suggested the intensive aggregation of plant-derived material and soil minerals promoted by the digestion and excretion of the earthworms (Don et al., 2008). The slightly lower thermal stabilities of PHLM (no release peak at 450°C), LIPID, and ALKY compounds (Fig. 5) in the OM of till-derived earthworm burrows indicated that these OM compounds were less aggregated or associated with soil minerals than found for OM of burrows from the loess-derived Luvisol.

With respect to the proportions of Py-FIMS compound classes, the OM composition of earthworm burrow samples from the till-derived Bt-horizon was similar to the OM composition of Luvisol topsoil samples from NE Germany published by Thiele-Bruhn et al. (2014; Fig. 6; $r = 0.91$). Other samples from the till-derived Bt-horizon were rather different to this topsoil-OM ($r = 0.38\text{--}0.74$). Clear differences between the cited topsoil data and the till-derived earthworm burrows [EB(T)] were found only in the proportions of PEPTI [higher for EB(T)] and PHLM [lower for EB(T)]. For the loess samples, such similarities were also present but to a smaller extent as compared to the till samples (not shown). The similarities underlined that the OM composition of earthworm burrow walls was strongly influenced by incorporated OM from the soil surface and the topsoil horizon.

4.3 Crack coatings and pinhole fillings

In comparison to earthworm burrows and uncoated cracks, thick crack coatings and pinhole fillings were dominated by heterocyclic N (NCOMP), benzonitrile, and naphthalene (Fig. 4), corresponding to increased proportions of C=O, C=C, C–N, and N–H functional groups (Fig. 2; $\text{WN } 1,641$, $1,624$, $1,616$, and $1,605\text{ cm}^{-1}$). The latter suggested the accumulation of carboxylic acids, aldehydes, and amides, which are characterized by an intermediate chemical stability (Demyan et al., 2012). The NCOMP proportions were larger (i.e., for loess-derived thick crack coatings, pinholes, and root channels, and till-derived pinholes) or within the range (all other samples) of NCOMP values observed by Py-FIMS for fine particle size fractions of a loamy sand (Schulten and Leinweber, 1991). The present NCOMP proportions were

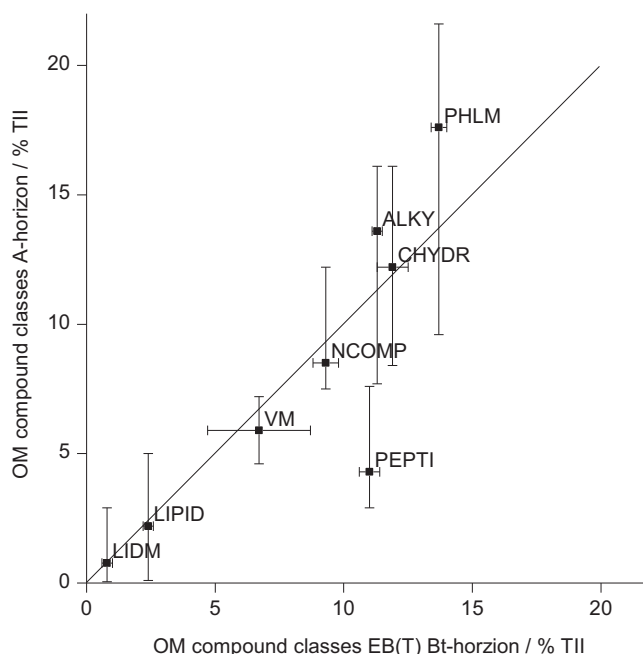


Figure 6: Organic matter (OM) composition of the earthworm burrow from the till-derived Luvisol Bt-horizon [sample EB(T)] related to the OM composition of a Luvisol A-horizon (Thiele-Bruhn et al., 2014). Error bars denote the standard deviations (x-axis) and minimum and maximum values (y-axis). The diagonal is the 1:1 line.

also larger than those reported for corresponding size-fractions from a loamy marl soil during a 34-y soil OM formation experiment (Schulten et al., 1992), a plausible difference that reflects the formation and stabilization of NCOMP as result of long-term humification processes during soil genesis.

The significant correlations between DRIFT signals of O–H groups from clay minerals at WN 3,690 and 3,616 cm^{-1} as well as of Si–O groups from clay minerals and quartz (WN 1,094, 1,031, 918, and 693 cm^{-1}) and the Py-FIMS class of NCOMP (Table 3) suggested that heterocyclic N-compounds were enriched in OM which is stabilized by soil minerals (Schulten et al., 1992); studies of Schulten and Leinweber (1999; 2000) showed that heterocyclic N-compounds were stabilized by clay minerals. In case of the till samples, higher NCOMP releasing temperatures of pinholes, compared to those of earthworm burrows, indicate a certain destabilization of NCOMP in the latter and/or a more strongly association of NCOMP with the illuviated clay particles in the pinholes. In these micro-environments once formed stable OM constituents probably are separated from biota-driven remobilization processes that are likely to occur in the earthworm burrows.

Heterocyclic N-compounds as well as Bn+Na, the latter found to be extremely increased in the till-derived pinholes, may also result from the incomplete combustion of OM (Kiersch et al., 2012b), which can originate for instance from burning of fossil fuels and/or vegetation fires. This suspicion of burning residues could be substantiated by Py-FIMS data from till samples that contained intensive signals of C_{15} to C_{17} alkanes at m/z 212, 226, and 240, i.e., from long-chain alkenes that are main components of diesel and diesel exhaust particulates (Liang et al., 2005). Although detected in samples from both Luvisols, the signals were prominent only for the till-derived Luvisol. In addition, release peaks of LIPID, ALKY, and PHLM below temperatures of 100°C indicated increased proportions of volatile organic compounds in the OM of till-derived uncoated cracks and crack coatings (Fig. 5). The till site was located relatively close (i.e., about 100 m) to a former railway line that was operated with diesel engines. The railway exhausts may have accounted for substantial amounts of diesel combustion particles in addition to the input of diesel exhaust residues from tractors, the latter is relevant for both sites.

5 Conclusions

The results from both, Fourier transform infrared spectroscopy in diffuse reflectance mode (DRIFT) and pyrolysis-field ionization mass spectrometry (Py-FIMS) indicate similar differences in the OM composition of biopore and crack surfaces from the Bt-horizon of Luvisols developed from loess and glacial till. The comparison of the data obtained by DRIFT and Py-FIMS suggests that the two methods can complement each other. The spectral information on O–H and Si–O groups of soil minerals and on O–H groups of free water and OM obtained from DRIFT can improve the interpretation of the release of volatile matter and low-molecular weight compounds obtained from Py-FIMS. The information of Py-FI mass spectra and thermograms can improve the assignment of DRIFT bands to OM functional groups and OM com-

pounds, in particular in DRIFT spectral regions of overlapping infrared signal intensities. The relation between bands from C=O and C=C bonds in the infrared range between WN 1641 and 1605 cm^{-1} and heterocyclic N-compounds, benzonitrile, and naphthalene enables to characterize soil samples with respect to these highly stable OM components by means of DRIFT spectroscopy. Information on the OM functional groups, compound classes, and thermal stabilities suggest that the OM composition of earthworm burrow walls from the Bt-horizon of both Luvisols is dominated by aggregated, chemically labile aliphatic-C and relatively stable components (i.e., lignin and alkylaromatics) originating from recent inputs by plant roots and earthworms. In contrast, the OM composition of crack coatings and pinhole fillings of Bt-horizons is enriched in chemically stable OM components (heterocyclic N-compounds, benzonitrile, and naphthalene). We conclude that OM is stabilized at these structural surfaces by clay minerals and that consequently the spatial distribution of chemically stable OM components in Luvisol Bt-horizons corresponds to the local mm- to cm-scale distribution of clay-rich crack coatings and pinhole fillings. As a future step, the OM composition at intact soil sample surfaces could be characterized in a more detailed way, when DRIFT spectra from undisturbed and/or intact samples are related to Py-FIMS data. The combination of the two methods seems promising to generate a more detailed mm-scale spatial distribution of OM-related physico-chemical properties such as sorption and wettability along the surfaces of preferential flow paths in structured soils.

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